HARDIK SRIVASTAVA

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EDUCATION

University of Washington — Seattle, WA
Master of Science in Data Science (GPA - 3.9/4) — Sept 2025 - Mar 2027

SRM Institute of Science and Technology — Chennai, TN
Bachelor of Technology in Computer Science with specialization in Big Data Analytics (GPA - 3.82/4) — May 2019 - Jun 2023

WORK EXPERIENCE

Graduate Research Assistant (P.I. - Prof. Zaid Harchaoui) — Sept 2025 - Present
Paul G. Allen School of Computer Science, University of Washington — *Seattle, WA*

- Architected a **post-training optimization** system for authorship obfuscation by fine-tuning a multi-adapter LoRA ensemble (Qwen2.5-32B) for dynamic style transfer; built metric-driven agents to guide **alignment** via constrained decoding, improving privacy by **40%**.
- Developed a **gradient-free few-shot optimizer** using in-context learning to dynamically weight 16-adapter ensembles, reducing author-specific configuration time from days to minutes. Outperformed grid-search baselines by **23%** in privacy-utility Pareto efficiency.
- Productionized the **agentic workflow** on **HPC clusters** utilizing Mistral-24B for robust evaluation and policy refinement. Engineered multi-objective feedback loops to enable **real-time policy iteration**, maintaining **88%** semantic similarity and **5%** perplexity drift.

Applied Scientist 2 — Jun 2023 - Aug 2025
JPMorganChase & Co. — *Hyderabad, TS*

- Developed a custom **cost-sensitive Focal Loss objective** for XGBoost & CatBoost to penalize false negatives; retrained production models to secure a **6%** recall boost and **$220M+** in annual savings while utilizing TreeSHAP to maintain strict feature interpretability.
- Productionized semi-supervised Graph Neural Networks across 60M+ transaction nodes, introducing **topology-aware edge sampling** and dynamic **neighborhood aggregation** to scale **fraud-detection** to 2M+ transactions/day with a **9.2%** precision boost.
- Architected a GPU-served **contrastive multimodal Transformer** (ViT + BERT-style decoder) for OCR-based check fraud detection, serving **500K+** requests/day. Improved fraud precision by **19%** while reducing latency via batched mixed-precision inference.
- Built a **Learning-to-Rank** information retrieval system over **3B+** KYC records using Pairwise Ranking loss to flag fraud entities; Outperformed the existing TF-IDF baselines, cutting **62% false positives** and saving **700+** analyst review hours daily in production.
- Engineered highly-scalable ML platforms and **distributed ETL pipelines** in AWS Glue for **terabyte-scale** financial data. Developed automated feature selection workflows that boosted downstream fraud capture by **14%**, reducing manual feature engineering overhead.

Research Intern — Jun 2022 - Sept 2022
McGill University - *Mitacs Scholar* — *Montreal, QC*

- Engineered a dense retrieval system for predictive **vocabulary generation** by training DeBERTa to encode image captions and vocab into a shared latent space; implemented **FAISS** for efficient k-nearest neighbor retrieval, outperforming the baseline by **6.9 MAP**.

Research Intern — Jan 2022 - Jun 2022
Samsung Research — *Bangalore, KN*

- Optimized production-scale **NLU intent routing** for Samsung Bixby via neural graph matching, reducing false positives by **12%**; Deployed INT8 inference via **Quantization-Aware** training for word disambiguation models, surpassing FP32 baselines by **2.9%** F1.

PROJECTS

Recursive Episodic Memory RAG: [code] Built a **memory compression** RAG framework for long-context LLM agents using recursive episodic memory consolidation, FAISS-based hierarchical retrieval, and adaptive replay. Enabled **persistent multi-session reasoning** leading to **41% context compression** under constrained token budgets via memory-aware reconstruction and recursive summarization.

PokerFace-RL [code] Built a multi-agent reinforcement learning framework for **strategic deception in LLMs** under asymmetric-information dialogue, modeling interaction as a POMDP and training policies using **PPO**, **GAE** & **QLoRA**. Introduced a two-stage opponent-masked reasoning architecture within a custom Gymnasium env. Raised deception success by **13 pp** over zero-shot baselines.

Agentic Ad Intelligence Engine [code] Architected a **multi-agent orchestration** pipeline for X (Twitter) using **Grok 4.1** to generate 100+ dynamic user personas and personalized ad copy, achieving **sub-5-second** end-to-end latency. Engineered a dynamic routing mechanism that ranked and delivered ads based on live engagement feedback, which increased click-through rates by **16%** in pilot campaigns.

SELECTED PUBLICATIONS

- Srivastava, H. **CM-BERT:** Multi-Modal Sentiment Analysis Using Text & Audio for Customer Support Centers (2023)
- Srivastava, H. **StyleLM:** Neural Text Style Transfer with Custom Language Styles for Personalized Communication Systems (2023)
- Srivastava, H. Using NLP Techniques for Enhancing Augmentative and Alternative Communication Applications (2021)

SKILLS

CS/ML	C++, Python, Java, SQL, PyTorch, TensorFlow, Sklearn, Spark, CUDA, LangGraph, HuggingFace, Generative AI
Infra/MLOps	AWS, Kubernetes, LangChain, MLFLow, RAG, Distributed Systems, Feature Stores, Vector Search